

Aly Ahmed

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science, Mechatronics Engineering, Honours, Artificial Intelligence Option

Apr 2027

- Relevant Courses
 - * MTE544 Autonomous Mobile Robots, ECE457B Fundamentals of Computational Intelligence, MTE484 Control Applications, MTE241 Introduction to Computer Structures and Real-Time Systems, MTE325 Microprocessor Systems and Interfacing for Mechatronics Engineering

EXPERIENCE

Robotics Software Engineer

Burnaby, BC

Red Rabbit Robotics

Jan 2026 - Aug 2026

- Contributed core software across an entire robot platform built from scratch, including motion primitives, teleoperation, vision software, motor controllers, and calibration tools.
 - * Engineered a C++/Python and ROS2 SDK for QDD actuators using SocketCAN to enable arm control
 - * Developed a full vision-to-placement pipeline that maps 2D image coordinates to 3D points using Orbbec RGBD depth and colour data and developed a camera-to-robot calibration routine via ChArUco boards.
 - * Collaborated on the development of functionality that enable multi-modal data collection pipeline, and executed many teleoperation sessions to capture data for VLA models.
 - * Created a custom robotic arm PID tuner using Bayesian Optimization and the Optuna library
- Contributed to software architecture discussions and helped make key decisions.
- Stress-tested robots on various tasks to evaluate real-world performance and provide feedback on any issues.
- Conducted performance benchmarking on hardware running inference, camera streaming, and control algorithms to select optimal hardware platform for the robot.
- Conducted deep-dive research and read papers to contribute to formulating the engineering team's direction.
- Performed mechanical assembly and electrical wiring, soldering, and crimping for robotic arms and control boxes.

Robotics Perception and Control R&D

Waterloo, ON

Robotic Interaction, Perception and Learning lab – University of Waterloo

May 2025 – August 2025

- Designed a probabilistic estimation pipeline for modeling and predicting the motion of flexible objects to support robot manipulation, implemented with factor graphs (GTSAM), ROS2, and Eigen. (C++)
- Implemented cartesian and joint impedance control with time-optimal trajectory tracking on the Franka Emika Panda, utilizing ROS2-Control (C++, Python)

Robotics Research Assistant

Waterloo, ON

Active & Interactive Robotics Lab – University of Waterloo

Sep 2025 – Dec 2025

- Explored methods for implementing impedance control on the TIAGo Pro from Pal Robotics

Platform Software Developer

Kanata, ON

Ford Motor Company

Sep 2024 – Dec 2024

- Executed embedded cryptography software improvements, successfully resolving 1000+ code issues and developed unit tests utilizing Google Test (C++)

Software Test Developer

Kanata, ON

BlackBerry QNX

Jan 2024 – Apr 2024

- Developed automated tests, and performed manual tests on Qualcomm, Bosch, and NXP ECUs, running QNX, and android automotive, producing invaluable engineering data, facilitating successful OEM demos
- Supported proof of concept demos to major BlackBerry customers like Nissan, and Ford through integration, development, and testing

SKILLS

Languages: C, C++, Python, Bash

Frameworks & OS: ROS/ROS2, FreeRTOS, QNX, Linux, OpenCV, GTSAM

Tools & Architecture: Git, Docker, AWS, Networking (TCP, UDP, CAN, SPI, I2C, UART)

Hardware & CAD: Altium Designer, SolidWorks, Fusion 360, Oscilloscopes, Soldering, Wire Harnessing